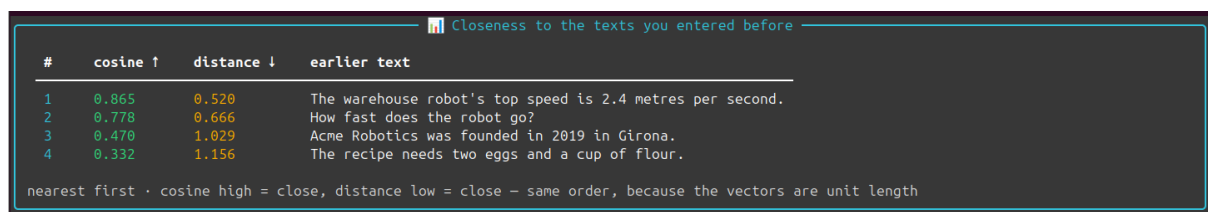


Play with the Embeddings Explorer

After using the /seed command, I entered the question "How fast is the warehouse robot?", which is very similar to asking how fast the robot is and is semantically related to the question about the warehouse robot's speed. As we can see in the image below, the embeddings have been generated correctly: the closest vectors correspond to the queries that are the most semantically related.

In contrast, the second-to-last result discusses robotics, so it may have a slight semantic connection. However, it is still relatively far away in the embedding space. The last result is completely unrelated, which is why it appears last and is located much farther from the query.



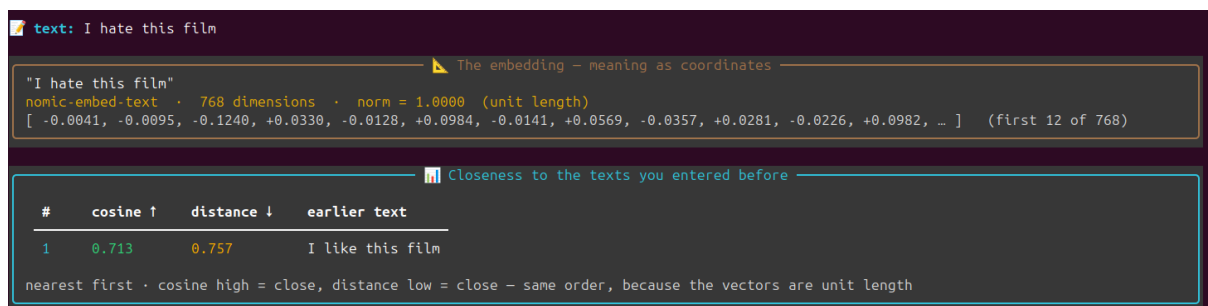
Closeness to the texts you entered before

#	cosine ↑	distance ↓	earlier text
1	0.865	0.520	The warehouse robot's top speed is 2.4 metres per second.
2	0.778	0.666	How fast does the robot go?
3	0.470	1.029	Acme Robotics was founded in 2019 in Girona.
4	0.332	1.156	The recipe needs two eggs and a cup of flour.

nearest first · cosine high = close, distance low = close – same order, because the vectors are unit length

Both metrics agree in order because the models generate unit vectors (of equal length), which means that a smaller angle (cosine) is always mathematically equivalent to a shorter straight-line (Euclidean) distance.

Testing with opposite expressions shows that the embeddings capture semantic meaning, the relationship between words, but not polarity or sentiment.



text: I hate this film

The embedding – meaning as coordinates

"I hate this film"

nomic-embed-text · 768 dimensions · norm = 1.0000 (unit length)

[-0.0041, -0.0095, -0.1240, +0.0330, -0.0128, +0.0984, -0.0141, +0.0569, -0.0357, +0.0281, -0.0226, +0.0982, ...] (first 12 of 768)

Closeness to the texts you entered before

#	cosine ↑	distance ↓	earlier text
1	0.713	0.757	I like this film

nearest first · cosine high = close, distance low = close – same order, because the vectors are unit length

When I entered several reviews, I was surprised to find that, although the model correctly matched two positive opinions initially, the completely opposite reviews remained mathematically very close; this shows me that the embeddings strongly prioritize the thematic context (talking about movies) and, although they capture some polarity, they don't give as much weight to emotional sentiment.

```
text: I would watch this movie a thousand times.

The embedding - meaning as coordinates
"I would watch this movie a thousand times."
nomic-embed-text · 768 dimensions · norm = 1.0000 (unit length)
[ +0.0232, +0.0339, -0.1747, +0.0261, -0.0164, -0.0071, -0.0298, +0.0962, +0.0044, +0.0245, -0.0654, +0.0485, ... ] (first 12 of 768)

Closeness to the texts you entered before

# cosine ↑ distance ↓ earlier text
1 0.615 0.878 I love this film
2 0.575 0.922 This is the worst movie I've ever seen.
3 0.528 0.972 I hate this film

nearest first · cosine high = close, distance low = close - same order, because the vectors are unit length
```

Collections, ingestion, and the retrieval gate

It can be observed that the information is indeed persistent, as indicated in the header “persisted on disk at ./collections-store”. This means that unlike the previous program, which stored everything in RAM (volatile memory that is erased upon restart), this one stores it on the hard drive, which is persistent. This prevents the loss of uploaded information, and each time the program runs, it loads and writes to RAM.

Initially, I used a PDF of university notes. The problem is that, being in slide format, it doesn't detect titles correctly, and the second strategy only produced one chunk. Therefore, I will use a more standard document.

When ingesting a C programming manual, the character-based strategy produced 193 chunks, while the section-based strategy generated only one chunk. I observed that information retrieval was infinitely better with the character-based strategy. This occurs because, although the PDF has clear visual titles, it lacks internal structural tags. Since the converter fails to detect them, the section-based strategy fails and groups the entire manual into a single fragment unmanageable for the AI, demonstrating that character-based partitioning is a much more reliable backup strategy.

The exact threshold I found to block off-topic questions without affecting the actual question was approximately 0.5, as the most focused questions hovered around 0.6 and the least focused around 0.4. Therefore, 0.5 or 0.55 are values that effectively divide the affinity between correct and off-topic questions.

This value cannot be programmed as a universal constant for all applications because it belongs exclusively to this embedding model and this particular document. If the document type or mathematical model is changed, the scoring range will vary, and the threshold will need to be readjusted.